

EVs to Power India's Path to Zero Emissions

As India walks ahead towards achieving net-zero goals, the promotion and consumption of EVs will play a significant role

SINCE 1947, A multitude of leading western observers doubted the ability of India to survive. However, not only did we defy our sceptics in standing united and democratic to our core, but we also emerged as a global leader across aspects of multilateralism, science, technology, medicine and combating climate change.

India is the world's fastest-growing major economy, with a GDP growth rate of 7%, and is seen as a bright spot for global economic revival. The country is expected to continue this growth momentum to become the third-largest economy, with GDP of over \$5 trillion, in the next three-four years. India's 1.4 billion population has a median age of around 28 years, creating a unique demographic bounty that can drive both consumption and production, and become the world's engine for growth. India's growing prominence in the global stage was also evident in the historic 2023 G20 presidency, where India played a key role in setting the agenda for this decade, particularly in combating global climate change.

In the aftermath of the Covid-19 pandemic, there has been a renewed focus on the need to address global climate change. Carbon dioxide emissions are the primary cause of rising atmospheric temperatures, resulting in global climate change. As a result, most major nations have



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declared net zero emission target dates and have enacted regulations to promote or mandate zero emissions technologies, particularly in the transportation and energy sector.

This drive to zero emissions is strongly supported by the Indian government. The Prime Minister had made a historic commitment at the COP26 summit to reduce India's carbon intensity by 45% by 2030 and reach net zero by 2070. Currently, India has very low per capita carbon emissions, driven by low penetration of energy-intensive goods and vehicles. With a fast-growing economy and a growing aspirational middle class, there will be growing affordability for goods and services. For example, over the next five years, India will see an addition of over 20 million cars and trucks and over 80 million two-wheelers. Incremental emissions from these vehicles would pose a significant challenge if they all ran on fossil fuels.

According to global studies, electric vehicles (EVs) have zero tailpipe emissions and the lowest lifecycle carbon emissions across all available automotive powertrain technologies. With India's solar power capacity expected to quadruple to 293 GW by 2030, lifecycle greenhouse gas emissions of EVs are projected to be 30-63% lower than their

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